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PHAS0056: Practical Machine Learning for Physicists Mini-project Report

THE SEARCH FOR THE HIGGS BOSON AT THE LHC

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ABSTRACT

The $VH \rightarrow Vb\bar{b}$ channel is an important but challenging mode for studying the Higgs boson because of the large background contribution. In this study, a cut-based analysis and a neural-network classifier are compared using the expected sensitivity as the main figure of merit. The final cut-based selection gives $Z_{\text{cut}} = 1.897$, while the fully optimised neural network reaches $Z_{\text{NN}} = 2.305 \pm 0.020$, corresponding to a relative improvement of 21.5%. Further studies show that the sensitivity plateaus at about 40–60% of the available training sample, that m_{BB} is the most important input variable, and that the training uncertainty is $\sigma_{\text{train}} = 0.024$. The high-NN-score region is cleaner and more Higgs-like than the cut-based selection, larger distortions of m_{BB} reduce the sensitivity, and a reduced-space joint search shows that the main NN result remains competitive.

1. INTRODUCTION

The discovery of the Higgs boson at the Large Hadron Collider (LHC) confirmed the mechanism of electroweak symmetry breaking predicted by the Standard Model [1, 2]. In this picture, particles acquire mass through their interaction with the Higgs field, and the observation of the Higgs boson provides direct evidence for this process [1]. Since its discovery, attention has shifted towards measuring its properties and testing whether they agree with Standard Model predictions [3, 4].

One important production mode at the LHC is the associated production of a Higgs boson with a vector boson (VH production). In this channel, the Higgs boson predominantly decays into a pair of bottom quarks, $H \rightarrow b\bar{b}$, which has the largest branching ratio [5]. In this study, the $VH \rightarrow Vb\bar{b}$ channel is identified by distinguishing signal events from background events, with the aim of maximising the discovery sensitivity using both traditional and machine learning approaches.

This decay channel is experimentally challenging because of the large background contribution, particularly from processes such as vector boson production with additional jets [6]. These backgrounds can mimic the signal signature and make the Higgs contribution difficult to isolate. In contrast, the $H \rightarrow ZZ \rightarrow 4\ell$ channel provides a much cleaner signal, but with a significantly smaller branching ratio, as illustrated in Fig. 1.1 [1, 2, 5].

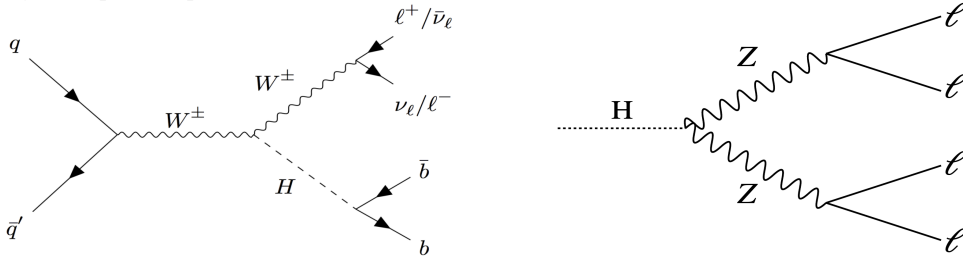


Figure 1.1: Representative Feynman diagrams for the $VH \rightarrow Vb\bar{b}$ signal process (left) and the cleaner $H \rightarrow ZZ \rightarrow 4\ell$ decay channel (right), illustrating the contrast between a high-background and a low-background Higgs decay mode.

To identify $H \rightarrow b\bar{b}$ events in the presence of large backgrounds, a set of kinematic observables is used to characterise differences between signal and background processes. The invariant mass of the two b -tagged jets, m_{bb} , provides a direct probe of the Higgs boson, with signal events expected to peak around 125 GeV. The angular separation between the two b -jets, $\Delta R(b_1, b_2)$, reflects the Higgs decay topology. The transverse momentum of the vector boson, p_T^V , is sensitive to the production kinematics of the VH system. The reconstructed top quark mass, m_{top} , is used to identify and suppress $t\bar{t}$ background contributions. The missing transverse energy, E_T^{miss} , provides information on neutrinos in the event and further helps to distinguish signal from background [6].

In high-energy physics, the separation of signal and background is usually formulated as a statistical test, where the background-only hypothesis is compared with the signal-plus-background hypothesis. The performance of an analysis is quantified in terms of a significance Z . For a counting experiment with signal

yield s and background yield b , the expected sensitivity is given by

$$Z = \sqrt{2 \left[(s + b) \ln \left(1 + \frac{s}{b} \right) - s \right]}, \quad (1.1)$$

which provides a more accurate estimate than the simpler $s/\sqrt{b}$ approximation, especially when the signal contribution is not negligible [7].

A traditional approach is to apply selection thresholds on physically motivated variables to define signal-enriched regions. This typically involves selecting events within a window of the di-jet invariant mass m_{bb} around the Higgs mass, together with additional cuts on other variables to suppress background contributions. Such methods are straightforward and interpretable, but they cannot fully retain the discrimination contained in the joint behaviour of several observables.

Machine learning provides an alternative approach. A neural network can take several input variables at once and build a non-linear discriminant between signal and background. This is useful in the $VH \rightarrow Vb\bar{b}$ channel, where m_{bb} is crucial but does not fully separate the signal from the background on its own, and the remaining discrimination is spread across other kinematic variables [8]. The final performance depends on the choice of inputs, the network architecture, and the training setup.

This study evaluates the performance of different strategies for identifying the $VH \rightarrow Vb\bar{b}$ signal and quantifies the improvement in sensitivity achieved using machine learning compared to a cut-based baseline. A neural network classifier is developed and optimised, and its performance is studied as a function of the training data and the choice of input variables.

2. METHODOLOGY

2.1. DATASET AND VARIABLES

The dataset used in this study consists of simulated proton–proton collision events for both signal and background processes. Each event is described by a set of reconstructed kinematic variables together with associated weights, which are used in the sensitivity calculation. Basic preparation of the dataset was carried out before the analysis. Columns without physical meaning, such as unnamed index columns, were removed, and the relevant variables and event weights were checked for consistency. The distributions of several key variables were then examined to confirm the expected differences between signal and background.

The separation between signal and background relies on a set of kinematic observables sensitive to the $VH \rightarrow Vb\bar{b}$ process. The invariant mass of the two b -tagged jets, m_{bb} , is the main discriminating variable, as signal events are expected to peak around the Higgs boson mass [6]. The angular separation between the two b -jets, $\Delta R(b_1, b_2)$, reflects the decay topology. The transverse momentum of the vector boson, p_T^V , is related to the production kinematics of the VH system. The reconstructed top quark mass, m_{top} , helps to identify and suppress background contributions from $t\bar{t}$ events. The missing transverse energy, E_T^{miss} , provides information on neutrinos in the event and contributes further discrimination.

Based on these considerations, the analysis is performed using the following set of input variables:

$$\{m_{bb}, \Delta R(b_1, b_2), E_T^{\text{miss}}, m_{\text{top}}, p_T^V\}. \quad (2.1)$$

These variables provide complementary information on the event kinematics and are used in both the cut-based and machine learning approaches.

The distributions of these variables for signal and background events are shown in Fig. 2.1. The variable m_{bb} shows a clear peak around the Higgs boson mass for signal events, while the background distribution is broader. Differences in shape are also seen in $\Delta R(b_1, b_2)$ and p_T^V , reflecting the different kinematic behaviour of signal and background. The variables m_{top} and E_T^{miss} provide additional sensitivity to background contributions, particularly from $t\bar{t}$ events [6].

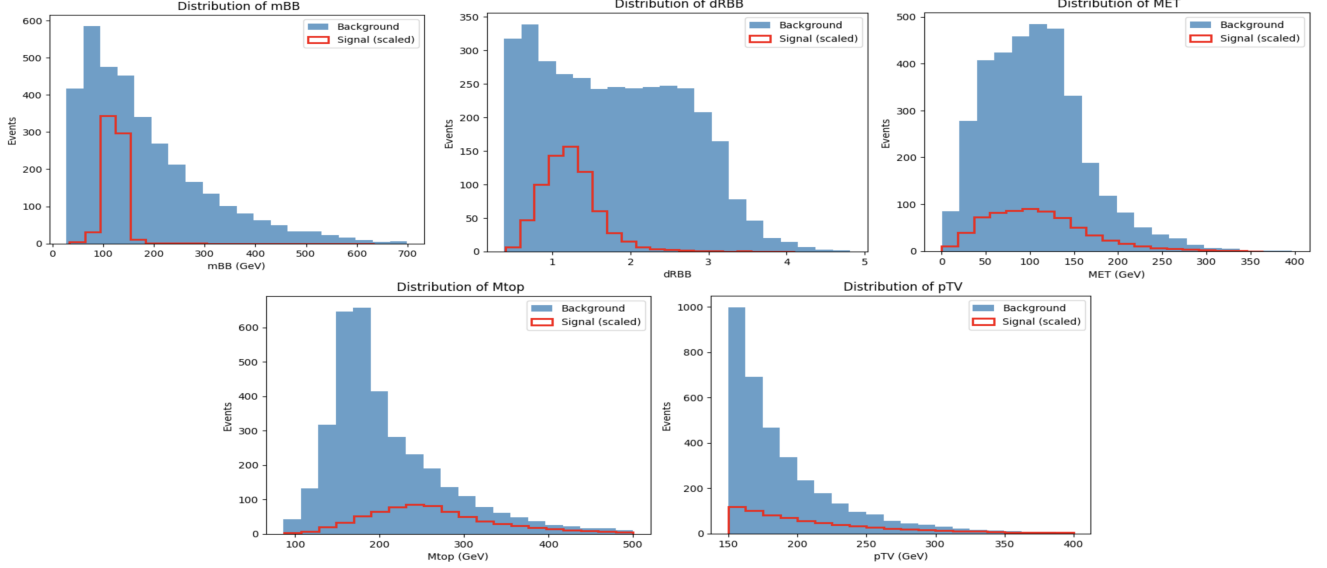


Figure 2.1: Distributions of the input variables for signal (red, scaled) and background (blue).

2.2. CUT-BASED ANALYSIS

A cut-based analysis is constructed as a baseline for comparison with the machine learning approach. The performance is evaluated using the expected sensitivity Z , as defined in Eq. (1.1) [7]. As a reference, the sensitivity is first computed on the full test dataset without any selection cuts.

The cut-based optimisation is performed sequentially using simple thresholds on a small set of physically motivated kinematic variables, while the di-jet invariant mass m_{bb} is retained as the final discriminant for the sensitivity calculation [6]. Starting from the baseline selection, candidate cuts are introduced one at a time and tested according to whether they improve the sensitivity.

The first variable scanned is the reconstructed top mass m_{top} , using thresholds of 190, 220, 250, 280 and 310 GeV. The subset passing the selected m_{top} requirement is then used to scan the angular separation of the two b -jets, $\Delta R(b_1, b_2)$, with thresholds of 1.0, 1.2, 1.5, 2.0 and 2.5. The same procedure is then applied to the transverse momentum of the vector boson candidate, p_T^V , with thresholds of 110, 130, 150, 170, and 190 GeV, and to the missing transverse energy, E_T^{miss} , with thresholds of 30, 40, 50, 60 and 70 GeV. At each stage, the scan is performed on the subset defined by the cuts already retained. A new cut is kept only if it improves the sensitivity relative to the current selection; otherwise it is discarded and the optimisation continues with the previous mask unchanged. The final cut-based selection is thus built only from thresholds that provide a genuine sequential improvement. The resulting sensitivity is then evaluated on the test dataset and used as the cut-based reference for comparison with the neural network approach.

The threshold scan and cut-based sensitivity are evaluated on the test dataset (VHbb_data_2jet_test.csv), consistent with the NN evaluation. Since the neural network is trained on the training sample and tuned on the validation sample, the test dataset provides an independent sample for its final evaluation. Using the same test dataset ensures that both approaches are compared on the same set of events.

2.3. NEURAL NETWORK ANALYSIS

2.3.1. Data Preparation and Sensitivity Evaluation

The selected kinematic variables defined in Section 2.1 are used as input features for the neural network [8]. These variables are scaled prior to training so that they lie within comparable numerical ranges. The same scaling is applied to the predefined training, validation, and test samples, allowing model performance to be monitored during training and reducing bias in the final evaluation. Event weights are retained throughout as they are required for a consistent calculation of the signal sensitivity. The neural-network output is treated as a continuous discriminant, and the signal and background yields are computed using event weights, with the sensitivity evaluation and selected analysis utilities implemented

with reference to the UCL support file `ucl_masterclass.py`.

An adaptive binning scheme is used to construct the distribution of the network output. The bin edges are determined through a transformation procedure that takes into account both statistical fluctuations and the distribution of event weights, avoiding sparsely populated bins and giving a more stable estimate of the sensitivity.

The expected significance is evaluated using a binned extension (`trafoD_with_error`) of the Asimov formula, where the contributions from individual bins are combined,

$$Z = \sqrt{\sum_i 2 \left[(s_i + b_i) \ln \left(1 + \frac{s_i}{b_i} \right) - s_i \right]}, \quad (2.2)$$

where s_i and b_i denote the signal and background yields in each bin of the neural network output [7]. The uncertainty on the sensitivity comes from the finite size of the Monte Carlo samples. Since the signal and background yields are estimated from weighted simulated events, their statistical uncertainties are determined by the sum of squared weights in each bin. The corresponding uncertainty on the sensitivity is obtained using standard error propagation,

$$\sigma_Z^2 = \sum_i \left(\frac{\partial Z}{\partial s_i} \sigma_{s_i} \right)^2 + \left(\frac{\partial Z}{\partial b_i} \sigma_{b_i} \right)^2. \quad (2.3)$$

This reflects the statistical fluctuation of the estimated sensitivity due to limited event statistics. Unlike the cut-based case, where the sensitivity is computed from a single signal region, the full shape of the network output is used in the NN-based evaluation [7].

2.3.2. Model Architecture and Baseline Training

A simple feed-forward neural network is used as a baseline for comparison with cut-based and optimised models. The network takes the selected kinematic variables as input and consists of a single hidden layer with 64 neurons using ReLU activation. The output layer has a single node with sigmoid activation, producing a continuous score that can be interpreted as the probability of an event being signal-like [9].

The model is kept simple so that it can serve as a baseline for the later optimisation studies. The implementation allows key hyperparameters, including the number of layers, hidden units, activation function, and optimiser, to be varied while the rest of the setup is held fixed. The baseline model is trained using binary cross-entropy loss with stochastic gradient descent (SGD) and a learning rate of 0.01. Training is performed for 30 epochs with a batch size of 256. Event weights are included during training to ensure that the optimisation reflects the weighted event distribution relevant for the sensitivity calculation.

The model is trained on the training sample, while the validation sample is used to monitor the training behaviour and check for possible overfitting. The evolution of the loss and accuracy is shown in Fig. 2.2, indicating stable convergence with no clear signs of overfitting.

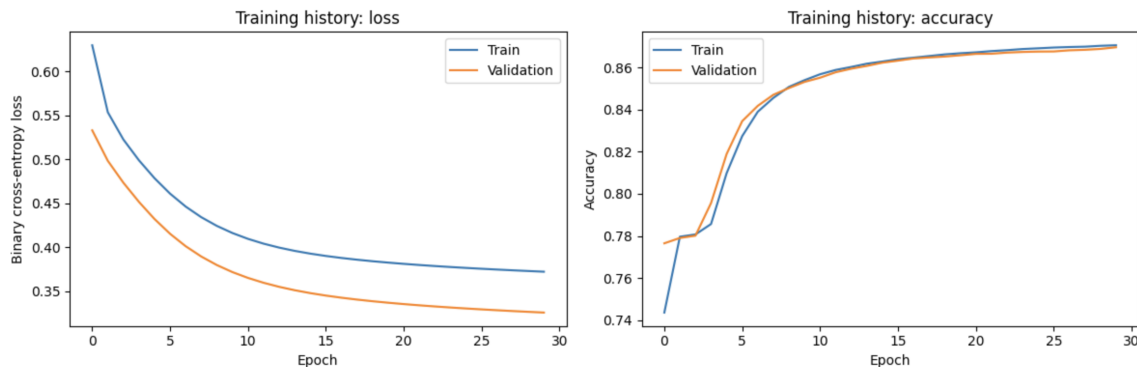


Figure 2.2: Training and validation loss and accuracy for the baseline neural network model.

After training, the model is applied to the test sample. The predicted output is stored as a continuous

discriminant, `decision_value`, which ranks events by their signal-like likelihood and is used to maximise the signal sensitivity defined in Section 2.3.1. The distribution of the network output is presented in the Results section.

2.3.3. Hyperparameter Optimisation

The optimisation starts from the baseline neural network defined above and is carried out using a sequential scanning procedure. One hyperparameter is varied in a loop while the others are kept fixed, and for each configuration the model is trained using the same pipeline and evaluated on the validation sample.

For each scan, candidate values are iterated over and the resulting sensitivities are recorded. The configurations are then ranked according to the validation sensitivity, and the best-performing value is selected and propagated to subsequent scans. This leads to a progressive refinement of the model, with each stage building directly on the previous configuration. The optimisation follows a fixed order. The number of hidden units is scanned first to determine the overall capacity of the network. This value is then fixed while the number of hidden layers is varied to study the effect of depth. With the architecture determined, the optimiser, activation function, and loss function are scanned sequentially.

The input scaling is handled separately from the model hyperparameters. At this stage, the network architecture and training setup are fixed to the best values obtained from the earlier scans. The dataset is then prepared using different scalers, and for each scaler the same model is retrained and evaluated on the validation sample. In this way, the effect of the scaling can be examined without changing the model configuration itself. After the best scaler is chosen, the dataset is reprocessed with this fixed transformation before the training-parameter scan.

All configurations in these scans are evaluated with the same training and evaluation pipeline, including event weights and the binned sensitivity calculation, so that the comparison is driven only by the parameter being scanned. The sequential scan is used here as an initial optimisation step under limited computing resources. Its main purpose is to identify a competitive region of the search space and to guide the reduced candidate ranges used later in the joint comparison between grid, stochastic, and Bayesian search.

2.3.4. Training Parameter Optimisation

After fixing the architecture and preprocessing, the training parameters are optimised using the same sequential procedure. Starting from the best hyperparameter configuration, the number of training epochs is scanned first to test where the performance begins to saturate, followed by the batch size which affects the training stability and then the learning rate to match the chosen optimiser.

At each step, only the parameter being tested is changed, while the others are kept fixed. The validation sensitivity is used to select the best setting, which is then carried forward to the next scan. After the training parameters have been fixed, a final model is trained with the selected configuration and evaluated once on the independent test sample, so that the reported sensitivity is based on data not used in the optimisation.

2.4. PERFORMANCE STUDY

2.4.1. Comparison of Analysis Strategies

To compare the performance of different approaches, the expected sensitivity is evaluated for a baseline selection without additional cuts, the optimised cut-based selection, the baseline neural network, and the fully optimised neural network. All sensitivities are computed using the same test dataset and evaluation procedure. The relative improvement is quantified by comparing the sensitivity of the neural network to that of the cut-based selection.

2.4.2. Training Statistics Study

Different fractions of the training dataset are used to test whether the available training statistics are sufficient. Signal and background events are resampled independently, while the validation and test samples are kept fixed. In the first scan, the signal training fraction is varied with the full background sample retained; in the second, the background training fraction is varied with the full signal sample retained. For each fraction, the same preprocessing and model configuration are used, and the sensitivity is evaluated on the

test sample. This allows the effect of limited signal and background statistics on the sensitivity to be studied separately, and shows whether the performance reaches a plateau as the training sample is increased.

2.4.3. Input Variable Importance

The importance of each input variable is studied with a leave-one-out test. Each variable is removed in turn from the feature training set, and the dataset is reprocessed using the same scaling as in the optimised model. For each reduced feature set, the network is retrained with the same architecture and training parameters as the full model, and the sensitivity is then evaluated on the test sample. The change in sensitivity relative to the full feature set is used to assess how much each variable contributes to the final performance.

2.4.4. Training uncertainty

To quantify the stability of the neural network training, the variability of the sensitivity under repeated trainings is evaluated using a fixed model configuration.

In this study, the same optimised hyperparameters and training parameters are used, while the model is retrained multiple times with different random initialisations. This is implemented by defining a separate training function (`fit_and_evaluate_nn_unseeded`) in which no fixed random seed is set internally. Instead, a different seed is applied before each training run, $\text{seed} = 42 + i$, where i labels the training iteration. For each run, the full training pipeline is repeated, including model construction, training with event weights, and evaluation on the test dataset using the same binned calculation. The resulting sensitivities are stored and used to estimate the spread of the model performance.

The training uncertainty is then defined as the standard deviation of the sensitivities over N independent runs,

$$\sigma_{\text{train}} = \sqrt{\frac{1}{N} \sum_{i=1}^N (Z_i - \bar{Z})^2}, \quad (2.4)$$

where Z_i is the sensitivity obtained in the i -th training and $\bar{Z}$ is the mean sensitivity. A visualisation is produced to display the run-to-run variation in sensitivity.

2.4.5. High NN output region

To investigate which events are selected by the neural network in its most sensitive region, a high-score region is defined from the model output. The neural network output (`decision_value`) is used as a discriminant, and a threshold is set using the 0.9 quantile of its distribution on the test dataset. Events above this threshold are taken as the high-NN-score region. The cut-based selected region is defined using the final optimised thresholds from Section 2.2. Both regions are constructed on the same test dataset.

The properties of the selected events are compared using the distributions of the input variables in three samples: the full test dataset, the high-NN-score region, and the cut-based selected region. Where needed, variables are converted to consistent units, and the corresponding mean values are recorded. In addition to the raw event counts, the weighted signal yield s_w and weighted background yield b_w are computed in each region using the event weights. A weighted purity is then defined as $P_w = \frac{s_w}{s_w + b_w}$, which measures how signal-enriched the selected region is.

2.4.6. Performance under distribution distortion

To study the effect of mismodelling in the most important input distribution, the variable m_{BB} is distorted in the test dataset. A range of distortions, $\pm 2.5\%$, $\pm 5.0\%$, and $\pm 10\%$ around the nominal value, is considered. This is done by multiplying m_{BB} in the test dataset by a scale factor, so that the rescaled invariant mass is shifted while all other variables remain unchanged.

The trained model obtained in Section 2.3 is then evaluated on each distorted dataset without retraining. The NN output is computed for each case, and the sensitivity is evaluated using the same binned procedure as in the nominal analysis. The distortion is applied only at evaluation stage, isolating the effect of mismodelling from changes in the training procedure.

3. RESULTS

3.1. CUT-BASED ANALYSIS AND OPTIMISATION

The sensitivity on the full test dataset, before any additional selection cuts are applied, is $Z_0 = 1.496$. This is taken as the baseline for the cut-based optimisation.

Among the variables scanned, the reconstructed top mass gives the largest improvement. Requiring $m_{\text{top}} > 220$ GeV increases the sensitivity to $Z = 1.896$. Applying an additional requirement of $\Delta R(b_1, b_2) < 2.0$ gives a small further improvement to $Z = 1.897$, which is the highest value obtained in the cut scan. The scans of p_T^V and E_T^{miss} do not lead to any further gain beyond this point, so they are not retained in the final selection. The final cut-based region is therefore defined by $m_{\text{top}} > 220$ GeV and $\Delta R(b_1, b_2) < 2.0$, together with the optimised m_{BB} window. Applied to the test dataset, the final cut-based selection gives

$$Z_{\text{cut}} = 1.897,$$

corresponding to an improvement of 26.8% over the no-cut baseline. After the final selection, 9488 events remain out of 16538 in the test sample. Most of the improvement comes from the m_{top} requirement, while the $\Delta R(b_1, b_2)$ cut gives only a small refinement. This optimised cut-based selection is used as the reference for the neural network comparison.

3.2. NEURAL NETWORK ANALYSIS

3.2.1. Baseline neural network performance

The baseline neural network was evaluated on the test sample before carrying out the full optimisation. Applied to the test dataset, it gives a sensitivity of

$$Z_{NN_{\text{base}}} = 2.196 \pm 0.020.$$

This is already higher than the optimised cut-based result from the previous section, showing that even a simple neural network can make use of information that is not fully captured by a sequence of one-dimensional cuts. The corresponding NN output distribution is shown in Fig. 3.1.

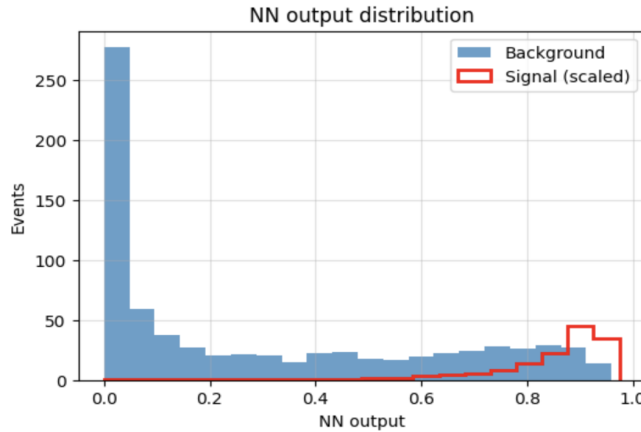


Figure 3.1: Output distribution of the baseline neural network for signal (red, scaled) and background (blue) events in the test sample.

Background events are concentrated at low output values, with a large population close to zero, while signal events are shifted towards the high-score region and accumulate near 1. This means that the network has learned a useful separation between the two classes: events with small output values are mainly background-like, whereas events with large output values are more likely to be signal-like. The overlap between the two distributions shows that the classification is not perfect, but the overall shape already indicates that the baseline model is capturing the main discriminating features of the dataset.

3.2.2. Hyperparameter and training optimisation

Starting from the baseline network, the optimisation gives a clear improvement in performance on both the validation and test samples. The highest-performing configuration found in the sequential validation scans is summarised in Table 3.1. This corresponds to a deeper and wider network than the baseline model, together with a tuned training setup.

Table 3.1: Best settings selected from the validation scans. The validation sensitivity shown for each row is the best value obtained in the corresponding scan stage.

Parameter	Best value	Best val. sensitivity
Hidden units	256	1.0049
Hidden layers	6	1.0212
Optimiser	SGD	1.0212
Activation	ReLU	1.0212
Loss	Binary cross-entropy	1.0212
Scaler	Standard	1.0212
Epochs	30	1.0212
Batch size	128	1.0286
Learning rate	0.02	1.0732

Several of the scan outputs show that the optimisation is not driven equally by every parameter. The activation scan shows the clearest difference between the tested options. ReLU gives the highest validation sensitivity, $\tanh$ is slightly lower, and sigmoid performs substantially worse. The loss-function scan also favours binary cross-entropy over mean squared error. By contrast, some of the other scans show only small differences. Standard and min-max scaling give similar validation sensitivities, and the results at 30 and 50 epochs are also close. The batch-size scan shows a mild preference for 128, while the learning-rate scan gives the largest final gain, with the best result at 0.02.

Using the full optimised configuration selected from the validation scans, a final model is trained and evaluated once on the independent test dataset. The corresponding sensitivities are

$$Z_{\text{NN, val}} = 1.073 \pm 0.081, \quad Z_{\text{NN}} = 2.305 \pm 0.020.$$

Compared with the baseline neural network result of $Z_{\text{NN}_{\text{base}}} = 2.196 \pm 0.020$, this gives an improvement of 4.96%.

The final test result is consistent with the sequential optimisation strategy. The validation sensitivity generally increases as the sequential scans proceed, and the scans favour configurations with larger capacity, ReLU activation, binary cross-entropy loss, and a relatively aggressive learning rate. The retrained model with this combination performs better than the baseline network on the independent test sample. This suggests that the validation sensitivity provides a useful criterion for model selection in this analysis, even though the gain over the baseline network remains moderate rather than dramatic.

3.3. COMPARISON OF CUT-BASED AND NEURAL-NETWORK APPROACHES

The sensitivities obtained at different stages of the analysis, all evaluated on the same test dataset, are summarised in Fig. 3.2. The no-cut baseline gives $Z_0 = 1.496$, which increases to $Z_{\text{cut}} = 1.897$ after the cut-based optimisation. The baseline neural network already reaches $Z_{\text{NN}_{\text{base}}} = 2.196 \pm 0.020$, while the fully optimised network gives the highest value,

$$Z_{\text{NN}} = 2.305 \pm 0.020.$$

The relative improvement of the final neural network over the optimised cut-based result is

$$\Delta Z_{\text{rel}} = \frac{Z_{\text{NN}} - Z_{\text{cut}}}{Z_{\text{cut}}} \times 100\% = 21.5\%.$$

A clear improvement is already present at the baseline NN stage, since even the simplest network performs better than the optimised cut-based selection.

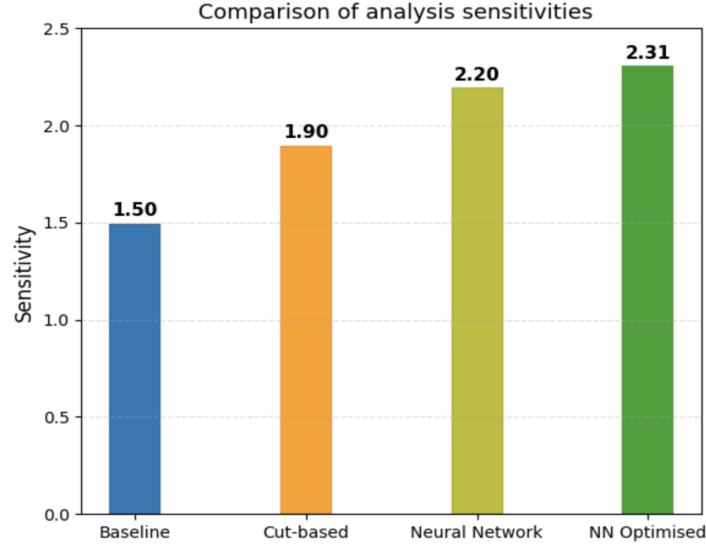


Figure 3.2: Comparison of sensitivities for the no-cut baseline, the optimised cut-based selection, the baseline neural network, and the fully optimised neural network.

The later optimisation gives a further increase in sensitivity. The change is smaller than the initial gain obtained by introducing the neural network itself, but it is still important for reaching the strongest final configuration.

The two approaches use the available information in different ways. The cut-based method is simple and easy to interpret, since each requirement can be linked directly to a physical feature of the event. The neural network is less transparent, but it combines all input variables into a single discriminant and makes use of correlations that are not captured by one-dimensional cuts. This is reflected in the higher final sensitivity. The optimisation outputs are also consistent with this picture. Validation loss and accuracy generally follow the same trend, but the ranking between configurations is most clearly reflected by the sensitivity itself. For this Higgs-search problem, sensitivity is the more relevant figure of merit, since it is directly related to how well the classifier enhances the signal over background.

In practical terms, the cut-based scan is faster and simpler, while the neural network requires repeated training and validation scans. For this dataset, however, the additional cost remains manageable and leads to a clear improvement in performance. The final optimised NN is therefore taken as the strongest classifier in this study.

3.4. PERFORMANCE STUDIES ON THE NEURAL NETWORK

3.4.1. Training statistics study

To test whether the available training statistics are sufficient, the signal and background training fractions were varied separately and evaluated on the test dataset. The results are shown in Fig. 3.3.

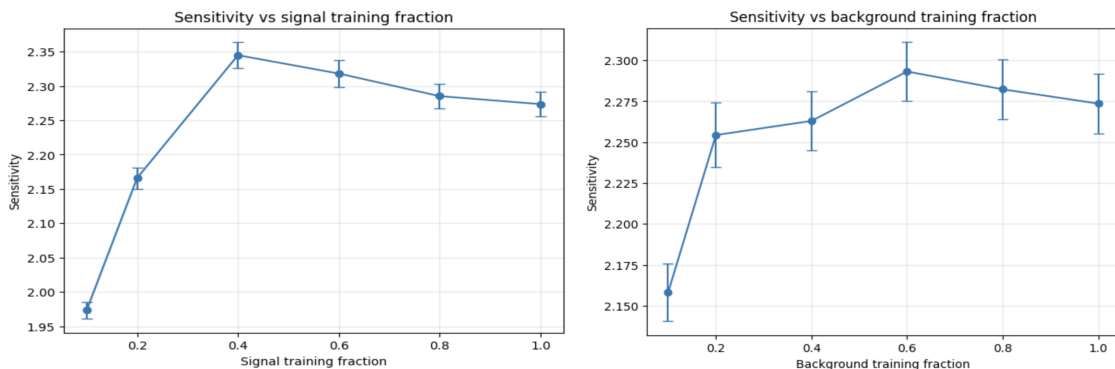


Figure 3.3: Sensitivity as a function of the signal training fraction (left), with all background events retained, and as a function of the background training fraction (right), with all signal events retained. Error bars show the statistical uncertainty on the sensitivity.

For the signal scan, the sensitivity increases strongly between 10% and 40%, rising from $Z = 1.973 \pm 0.012$ to $Z = 2.345 \pm 0.019$. Beyond 40%, the changes are smaller and the sensitivity decreases slightly, reaching $Z = 2.274 \pm 0.018$ at 100%. For the background scan, the sensitivity rises from $Z = 2.158 \pm 0.017$ at 10% to $Z = 2.254 \pm 0.020$ at 20%, and then changes only modestly, with the highest value $Z = 2.293 \pm 0.018$ obtained at 60%.

Both scans suggest that the model is not strongly limited by the full available training sample, although the exact behaviour is not identical. The signal scan reaches its highest value at about 40%, while the background scan changes only weakly once the fraction is above about 20%–60%. The validation loss and accuracy show the same pattern: the largest changes appear at low fractions, while later increases in sample size give only small differences. This suggests that the gain from additional training statistics has already become limited before 100% of the data is used.

3.4.2. Input variable importance

The effect of each input variable was examined by removing it in turn, while keeping the final network configuration fixed and evaluating the result on the same test dataset. The results are summarised in Table 3.2.

Table 3.2: Leave-one-out study of input variable importance. $Z_{NN} = 2.305 \pm 0.020$ as reference.

Removed feature	Sensitivity	Uncertainty ($\times 10^{-2}$)	Drop in Z	Relative drop
m_{BB}	1.976	1.44	0.329	14.3%
E_T^{miss}	2.205	1.58	0.100	4.4%
p_T^V	2.240	1.68	0.065	2.8%
m_{top}	2.262	1.97	0.043	1.9%
$\Delta R(b_1, b_2)$	2.280	1.82	0.025	1.1%

The largest degradation is seen when m_{BB} is removed, for which the sensitivity falls to $Z = 1.976 \pm 0.014$. This is a much larger drop than for any other variable and confirms that m_{BB} is the most important input to the classifier. This is physically reasonable, since the Higgs signal is directly associated with a resonance in the di-jet invariant mass.

Removing E_T^{miss} also gives a visible reduction in sensitivity, while removing p_T^V and m_{top} has a smaller effect. The weakest change is seen for $\Delta R(b_1, b_2)$, for which the sensitivity remains close to the baseline value. Overall, the network relies mostly on m_{BB} , while the other variables provide more limited supporting discrimination.

3.4.3. Training uncertainty

To estimate the repeated training variation, the final neural network was retrained 10 times with different random initialisations and evaluated on the same test dataset. The resulting sensitivities are shown in Fig. 3.4. The sensitivities lie in the range $2.240 \lesssim Z \lesssim 2.340$, with a mean value of

$$\bar{Z} = 2.286 \quad \text{and} \quad \sigma_{\text{train}} = 0.024.$$

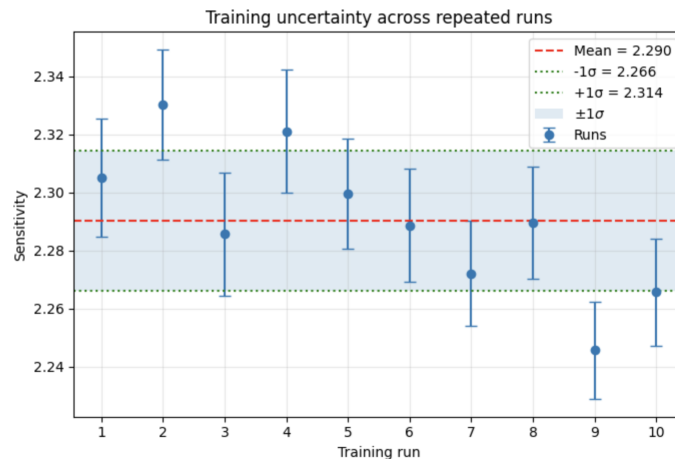


Figure 3.4: Sensitivity obtained from 10 repeated trainings of the final neural network configuration. The dashed red line shows the mean sensitivity, and the shaded band indicates the $\pm 1\sigma$ spread.

Most runs lie close to the mean, with a modest spread between trainings. The training uncertainty σ_{train} is slightly larger than the statistical uncertainty quoted for a single run, at about 1.2 times the final sensitivity uncertainty. However, it remains much smaller than the difference between the cut-based and neural-network sensitivities (≈ 0.4), which shows that the NN improvement is stable under repeated retraining.

3.4.4. High NN output region and comparison with the cut-based selection

The high-NN-score region was defined by selecting the top 10% of events in the NN output on the test dataset, corresponding to a threshold of `decision_value` = 0.9774. The final cut-based region was constructed on the same test dataset for direct comparison.

The high-NN-score region contains 1634 signal events and 20 background events, while the cut-based selection contains 10602 signal events and 3554 background events. The cut-based region keeps more signal events overall, but it also keeps far more background. This is also reflected in the weighted purity, which is 0.195 for the NN-selected region and 0.0169 for the cut-based selection.

The variable distributions are shown in Fig. 3.5. In all panels, blue denotes the full test sample, orange the cut-based region, and green the high-NN-score region. The clearest difference appears in m_{BB} : the NN-selected events form a much sharper peak around the Higgs mass, with a mean value of 124.1 GeV, while the cut-based region is broader and shifted lower, with a mean of 115.4 GeV. The NN-selected region also has smaller $\Delta R(b_1, b_2)$ and larger average values of E_T^{miss} , m_{top} , and p_T^V . The two selections overlap, but they are not selecting the same part of phase space. The high-score NN region is narrower and cleaner, and overall looks more Higgs-like than the cut-based selection.

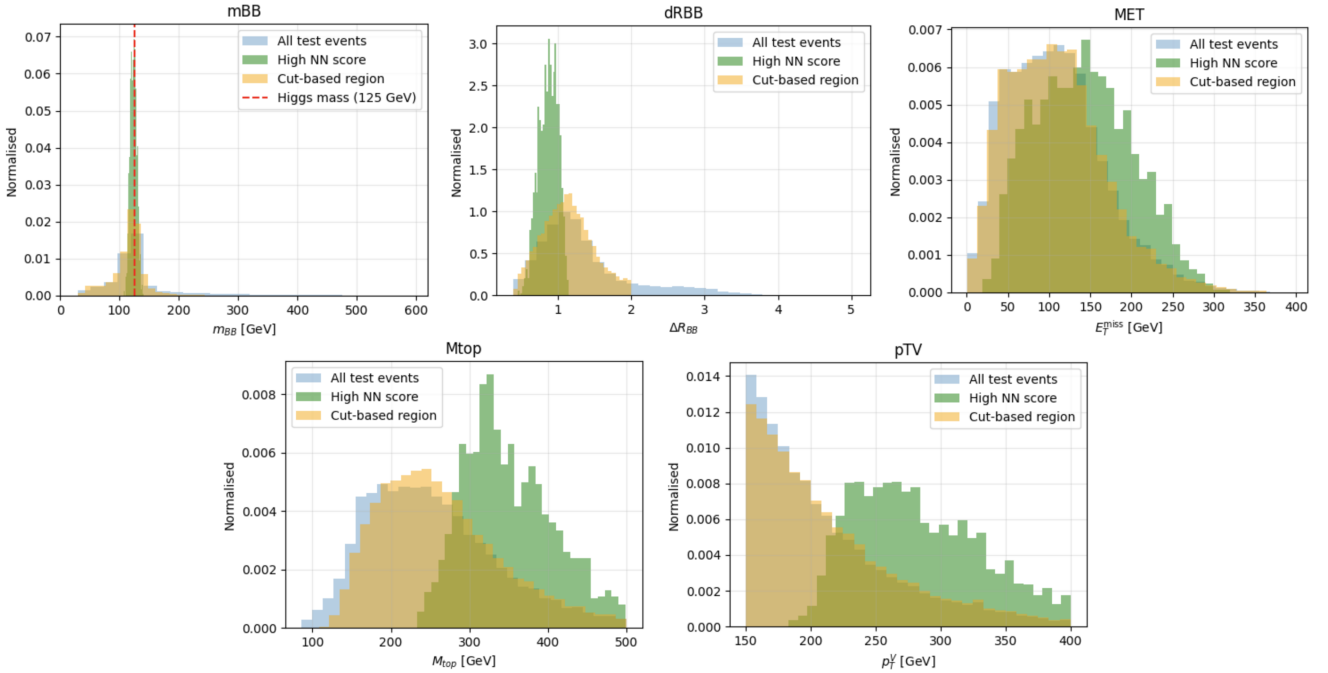


Figure 3.5: Distributions of the five input variables for the full test sample (blue), the cut-based selected region (orange), and the high-NN-score region (green).

3.4.5. Performance under distribution distortion

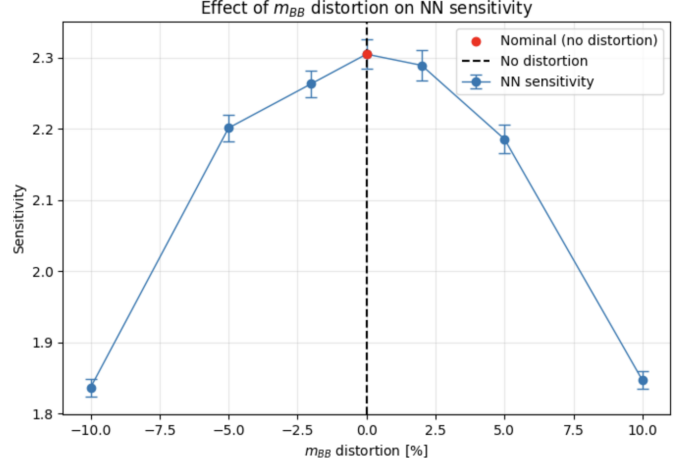
To mimic the effect of imperfect simulation, the final neural network was re-evaluated on the test sample with an artificially distorted m_{BB} distribution. The sensitivity is highest at the nominal point, with $Z = 2.305 \pm 0.020$. Small distortions of order $\pm 2\%$ have only a limited effect, while the degradation becomes clear at $\pm 5\%$. For the largest distortions, the sensitivity drops much more strongly, reaching $Z = 1.837 \pm 0.012$ at -10% and $Z = 1.847 \pm 0.012$ at $+10\%$. The full set of results is shown alongside the table and curve in Fig. 3.6.

This shows that the NN performance is reasonably stable against small mismodelling in m_{BB} , but not against larger shifts. Since the classifier relies strongly on m_{BB} , inaccurate modelling of the mass

distribution directly reduces the expected sensitivity. This is consistent with the earlier variable-importance study, where m_{BB} was found to be the dominant input variable.

Distortion mBB%	Sensitivity	Unc. ($\times 10^{-2}$)	Drop
-10	1.837	1.23	20.3%
-5	2.201	1.86	4.5%
-2	2.264	1.82	1.8%
0	2.305	2.03	0.0%
+2	2.289	2.14	0.7%
+5	2.186	1.97	5.2%
+10	1.847	1.25	19.9%

(a)



(b)

Figure 3.6: Effect of m_{BB} scale distortion on the neural network sensitivity. The table (a) summarises the sensitivity and relative drops, while the plot (b) shows the dependence of sensitivity on the distortion level.

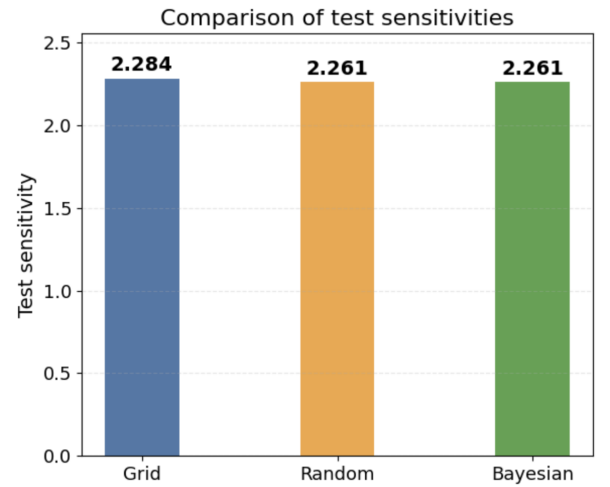
3.5. EXTENDED STUDIES: HYPERPARAMETER SEARCH STRATEGY COMPARISON BETWEEN GRID, STOCHASTIC AND BAYESIAN

Three hyperparameter search strategies were compared on the same reduced candidate space: grid search, stochastic search, and Bayesian search. The search space contained 432 candidate configurations, defined by combinations of hidden units, number of hidden layers, activation function, optimiser, learning rate, scaler, and batch size. This reduced space was chosen so that a joint comparison could be carried out within the available computing budget, and the ranges were guided by the earlier sequential scans. In all cases, configurations were trained and ranked using the validation sensitivity. Grid search evaluated the full reduced space, whereas stochastic and Bayesian search were each restricted to 30 evaluations [10].

The main results are summarised in Fig. 3.7. Grid search achieved the highest validation sensitivity, $Z_{\text{val}} = 1.133 \pm 0.082$, and also the highest test sensitivity, $Z_{\text{test}} = 2.284 \pm 0.021$. Random and Bayesian search both reached a lower best validation sensitivity of $Z_{\text{val}} = 1.092 \pm 0.083$, and both gave $Z_{\text{test}} = 2.261 \pm 0.020$. Within the reduced search space considered here, grid search gave the strongest result, while the final sequentially optimised model used in the main analysis remained competitive at test level, $Z_{\text{NN}} = 2.305 \pm 0.020$.

	Grid search	Random search	Bayesian search
N_{eval}	432	30	30
Space tested	1.000	0.069	0.069
Best Z_{val}	1.133 ± 0.082	1.092 ± 0.083	1.092 ± 0.083
Best Z_{test}	2.284 ± 0.021	2.261 ± 0.020	2.261 ± 0.020
Hidden units	64	64	64
Hidden layers	4	4	4
Activation	ReLU	ReLU	ReLU
Optimiser	Adam	Adam	Adam
Learning rate	0.01	0.02	0.02
Scaler	Standard	MinMax	MinMax
Batch size	128	256	256
Total time [s]	27459	1597	1484
Mean time per trial [s]	63.6	53.2	49.5

(a)



(b)

Figure 3.7: Comparison of the three hyperparameter search strategies. (a) Summary table of the three search strategies. (b) Comparison of the corresponding test sensitivities.

The difference in computational cost, however, is much larger than the difference in performance. Grid search evaluated all 432 configurations and required about 2.75×10^4 s in total. By contrast, random and

Bayesian search each tested only 30 configurations, corresponding to 6.94% of the full space, with total runtimes of about 1.60×10^3 s and 1.48×10^3 s respectively. Bayesian search was slightly faster than random search in this study, but did not produce a better final configuration, likely because the search space is discrete, low-dimensional, and sampled with a limited evaluation budget.

For the reduced joint comparison carried out here, grid search performed best among the three search strategies considered, because the candidate space was still small enough to be explored more completely and it gave the highest final sensitivity. Random and Bayesian search were much cheaper computationally, and would become more attractive for a larger search space, but in the present case their main advantage was reduced runtime rather than improved sensitivity.

4. CONCLUSION AND FUTURE OUTLOOK

This project has compared a cut-based analysis with a neural-network classifier for the $VH \rightarrow Vb\bar{b}$ channel. For the dataset studied here, the neural network gives the stronger result, improving the sensitivity from $Z_{\text{cut}} = 1.897$ to $Z_{\text{NN}} = 2.305 \pm 0.020$. The improvement does not come from one particular tuning choice alone. The main change comes from treating the problem as a multivariate classification task, while the later optimisation refines the model and improves the final result further.

The supporting studies are consistent with the physics of the channel. The dominant role of m_{BB} reflects the Higgs resonance structure in the $b\bar{b}$ final state, while the remaining variables provide additional separation through the event topology and production kinematics. The high-NN-score region follows the same picture, selecting a narrower and cleaner subset of events than the cut-based selection.

The robustness studies also place the result in context. The training-statistics scan shows that the model is already close to its best performance before the full training sample is used, and the repeated-training study shows that the retraining spread is small compared with the difference between the NN and cut-based sensitivities. The distortion study shows that the NN gain depends on modelling m_{BB} reasonably well, with only small changes under mild distortions and a clear loss in sensitivity for larger shifts.

In general, the cut-based analysis remains useful as a transparent reference, but the stronger performance comes from treating the problem with a multivariate classifier. In this channel, the gain does not come from one additional variable or one particular hyperparameter choice. It comes from combining the full set of observables and using their correlations to separate signal from background more effectively than a sequence of one-dimensional cuts.

There are, however, clear limitations in the present study. The analysis is based on simulated events, so the final performance depends on how well the simulation reproduces the relevant kinematic distributions. This is already visible in the m_{BB} distortion study, where the sensitivity degrades once the mass distribution is shifted away from the nominal shape. The hyperparameter optimisation also has limitations. The sequential scan used here is systematic and easy to interpret, but on its own it does not capture possible interactions between hyperparameters. In this project, it was used mainly to identify a promising region of the search space before carrying out a reduced joint comparison with grid, stochastic, and Bayesian search under the available computing budget. The final configuration should therefore be understood as the best one found within the reduced search space considered here, rather than a unique global optimum of the full hyperparameter problem.

A practical extension would be to move beyond a single binary classifier and treat the dominant background categories separately, in particular $V + b\bar{b}$, $t\bar{t}$, and diboson production. These backgrounds populate phase space differently, and the present binary setup compresses them into one inclusive class. A direct multi-class network, or a cascade in which one classifier first suppresses $t\bar{t}$ and a second refines the remaining separation, would test whether the residual overlap can be reduced further. The hyperparameter search could also be revisited for a larger model space, where exhaustive grid search would no longer be practical and comparisons with random or Bayesian search would need to be made under a matched evaluation budget.

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